

Control work 7, and work on the mistakes

The five abridged multiplication formulae, in both directions, in one paper.

LESSON 110–111

2 × 40 MIN

MATEMATIKA 7, NAZORAT ISHI 7

S8 2 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions

3 min

The paper

40 min

Answers

12 min

Diagnosis and rewrite

20 min

The map

5 min

LEARNING OBJECTIVES

• Expand squares and cubes of binomials accurately under time.

• Factorise by common factor, formula, grouping and splitting.

• Use the formulae for mental calculation.

• Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National

pp. 282–326

Cambridge

Stage 8, pp. 18–30

Workbook

Control paper M7

01

Explanation

The paper — 25 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Expand $(x + 6)^2$, $(2a - 5)^2$ and $(x - 1)^3$	5	L98–103
2	Factorise $x^2 - 49$ and $9a^2 - 16b^2$	4	L100–101
3	Factorise $2x^2 - 50$ completely	4	L106–107
4	Factorise $x^2 + 2x - 15$ and $x^3 + 8$	5	L104–107
5	Find 98^2 and $64 \cdot 56$ without a calculator	4	L108–109
6	Simplify $(a + b)^2 - (a - b)^2$	3	L108–109

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for the middle term of each square and one for the four signs of the cube; Q3 two for taking the 2 out before anything else; Q5 one for naming the formula used; Q6 two for expanding both brackets rather than subtracting them whole.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
middle term dropped	$(x + 6)^2 = x^2 + 36$	$x^2 + 12x + 36$
coefficient not squared	$(2a - 5)^2 = 2a^2 - 20a + 25$	$4a^2 - 20a + 25$
cube ended with a plus	$(x - 1)^3 = x^3 - 3x^2 + 3x + 1$	$\dots - 1$
sum of squares “factorised”	$x^2 + 49 = (x + 7)^2$	it does not factorise
common factor left in	$2(x^2 - 25)$	$2(x - 5)(x + 5)$
cube formula signs matched	$x^3 + 8 = (x + 2)(x^2 + 2x + 4)$	$(x + 2)(x^2 - 2x + 4)$
brackets subtracted whole	$(a + b)^2 - (a - b)^2 = (2b)^2$	expand both first

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

The chapter as one map

FORMULA	READ LEFT TO RIGHT	READ RIGHT TO LEFT
$(a \pm b)^2$	expanding	recognising a perfect square
$a^2 - b^2$	a quick product	the fastest factorisation
$(a \pm b)^3$	expanding	recognising a perfect cube
$a^3 \pm b^3$	checking	factorising
splitting the middle term	—	factorising a trinomial

LOOKING FORWARD

Every one of these formulae is about to earn its keep: algebraic fractions cancel only when numerator and denominator are factorised, and that is the next four lessons.

02 Worked examples

EXAMPLE 1 Model answer, Q1: expand $(2a - 5)^2$.

① $a = 2a, b = 5.$

② $(2a)^2 = 4a^2$

The coefficient is squared too.

③ $-2 \cdot 2a \cdot 5 = -20a$

④ $+25$, so the answer is $4a^2 - 20a + 25$.

ANSWER

$$4a^2 - 20a + 25$$

EXAMPLE 2 Model answer, Q3: factorise $2x^2 - 50$ completely.

① Common factor 2: $2(x^2 - 25)$.

② $x^2 - 25 = (x - 5)(x + 5)$

Do not stop at the previous line.

③ $= 2(x - 5)(x + 5)$

ANSWER

$$2(x - 5)(x + 5)$$

EXAMPLE 3 Model answer, Q5: find $64 \cdot 56$.

① Both are 4 from 60.

② $= (60 + 4)(60 - 4)$
A difference of squares.

③ $= 3600 - 16$

④ $= 3584$

ANSWER

3584

03 Interactive model

Hand back Q1 and Q2 together: expanding and factorising are the same five formulae read in opposite directions, and the paper is easier to learn from that way.

The chapter in twelve questions

$(a + b)^2$ equals:

$$a^2 + b^2$$

$$a^2 + 2ab + b^2$$

$$a^2 - 2ab + b^2$$

$$2a + 2b$$

$(a - b)^2$ ends in:

$$+ b^2$$

$$- b^2$$

$$+ b^3$$

$$- b$$

$(x + 6)^2$ equals:

$$x^2 + 36$$

$$x^2 + 12x + 36$$

$$x^2 + 6x + 36$$

$$x^2 + 12x + 6$$

$(2a - 5)^2$ begins with:

$(x - 1)^3$ ends in:

$x^2 - 49$ factorises to:

$x^2 + 49$ factorises to:

$2x^2 - 50$ factorises completely to:

$x^2 + 2x - 15$ factorises to:

$x^3 + 8$ factorises to:

98^2 equals:

$(a + b)^2 - (a - b)^2$ equals:

$4ab$

$2b^2$

$4b^2$

0

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Square of a sum	Yig'indining kvadrati	Квадрат суммы
Difference of two squares	Kvadratlar ayirmasi	Разность квадратов
Cube of a difference	Ayirmaning kubi	Куб разности
Sum of two cubes	Kublar yig'indisi	Сумма кубов
To factorise	Ko'paytuvchilarga ajratish	Разложить на множители
Completely	To'liq	Полностью
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 loses a mark most often through:

the first term

the middle term

the sign of b^2

the brackets

Q2 needs:

a common factor

a difference of squares

grouping

splitting

Q3 begins with:

the formula

the common factor

grouping

splitting

In Q4, $x^3 + 8$ has middle sign:

plus

minus

either

none

Q5 uses $64 \cdot 56$ as:

$(60 + 4)(60 - 4)$

$(64 - 8)^2$

$60 \cdot 60$

$(60 + 4)^2$

Q6 must begin by:

subtracting the brackets

expanding both brackets

factorising

substituting

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $(x + 6)^2$

ANSWER $x^2 + 12x + 36$

2. $(2a - 5)^2$

ANSWER $4a^2 - 20a + 25$

3. $(x - 1)^3$

ANSWER $x^3 - 3x^2 + 3x - 1$

4. $x^2 - 49$

ANSWER $(x - 7)(x + 7)$

5. $9a^2 - 16b^2$

ANSWER $(3a - 4b)(3a + 4b)$

6. 98^2

ANSWER 9604

7. $64 \cdot 56$

ANSWER 3584

Medium · the standard question**7 PROBLEMS**

1. $2x^2 - 50$

ANSWER $2(x - 5)(x + 5)$

2. $x^2 + 2x - 15$

ANSWER $(x + 5)(x - 3)$

3. $x^3 + 8$

ANSWER $(x + 2)(x^2 - 2x + 4)$

4. $(a + b)^2 - (a - b)^2$

ANSWER $4ab$

5. $3x^2 + 18x + 27$

ANSWER $3(x + 3)^2$

6. $x^2 - 7x + 12$

ANSWER $(x - 3)(x - 4)$

7. Does $x^2 + 49$ factorise?

ANSWER No

Hard · stretch and reason

7 PROBLEMS

1. $x^4 - 81$

ANSWER $(x - 3)(x + 3)(x^2 + 9)$

2. $2x^3 - 54$

ANSWER $2(x - 3)(x^2 + 3x + 9)$

3. $(3a + 1)^2 - (3a - 1)^2$

ANSWER $12a$

4. $x^2 + 6x + 9 - y^2$

ANSWER $(x + 3 - y)(x + 3 + y)$

5. $125^2 - 75^2$

ANSWER 10000

6. $a^3 - a$

ANSWER $a(a - 1)(a + 1)$

7. Simplify $\frac{x^2 - 36}{x - 6}$

ANSWER $x + 6$

07 Homework**Homework — 5 tasks**

Rewrite every question you lost a mark on in full, from the first line.

1. Rewrite in full every question on which you lost a mark.
2. Write out the five formulae with one worked example of each.
3. Factorise $3x^2 - 75$ and $x^3 - 27$ completely.
4. Find 106^2 and $73 \cdot 67$ by the formulae.

5. Simplify $(x + 4)^2 - (x - 4)^2$ and say which slip you were avoiding.